

Lone Survivor on PS3/PS Vita

The indie horror game now on PS3 and PS Vita. Console to get a Director's Cut <http://dgit.in/18jhjys>

PES 2014 demo rolls out

Immediately after FIFA 14 demo appeared, Konami released a free demo of PES 2014 for all platforms <http://dgit.in/1ezuUrT>

output as is done with normal USB headers. Only then can you truly call it capable of charging two devices simultaneously. Most power banks have more than one output but they have a single supply that is shared between the two outputs.

The USB standard specifies 500mA per port for USB 2.0 and 900mA per port for USB 3.0. However, manufacturers understand that charging a simple smartphone with a 2000mAh battery would take upwards of 4 hours, so in order to reduce that time these standards went for a toss. Smartphone chargers supply somewhere between 500mA to 1.5A which also causes the device and the charger to heat up considerably. Then there are certain devices like the iPad which requires 2.1A and not all chargers support this.

While this might seem a bit absurd the reasoning is that the iPad specific chargers have a few resistors hooked up on the data lines (Pins 2 and 3 of USB 2.0) which helps the device identify if the charger is certified by Apple. This is why even



Zebronics ZEB-PG7200 - value for money

if you do end up with a charger/power bank which can provide 2A or more it might not charge an iDevice.

Overcurrent Protection

Basically in electric circuits, if there is a short circuit or a ground fault then the current flow increases to the point that the components start overheating and then they burnout. An overcurrent protection circuit simply shuts down the circuit whenever it detects excess current flow. Over current protection is needed at both the output and the input ports of the power bank. Without this protection the risk lies not only in the power bank getting dam-

aged but even the device being charged. It is also often advertised as short-circuit protection.

Overtemperature protection

Most ICs which are at the core of the charging circuit feature an overtemperature protection mechanism with some having an extra sensor around the battery compartment. The moment the temperature inside the battery compartment crosses a threshold the cut-off mechanism is triggered.

Overcharge Protection

When a battery has reached its maximum capacity, charging it further leads to bloating and subsequently causes it to blow up. If you don't believe us then there are plenty of videos online(<http://dgit.in/19nK11B>).

Now you don't need to worry about overcharging your device because most modern smartphones and tablets have built in overcharge protection. Here it's important because you need to be assured that the power bank won't suffer the same fate as the video linked above. This



Portronics Charge X - packing more charge per cm3

way you can leave your powerbank to charge unsupervised without having to come back to charcoal.

Similar to overcharge protection is over discharge protection which stops the power bank from continuing to charge the device when the power bank nears the end of its capacity. Completely discharging and charging is somewhat bad for batteries and they may lose capacity over a long period as a result. Recovering from such a state usually involves charging at a really low current till full and that cannot be done without removing the battery from the power bank.

Build quality

Almost every powerbank we came across turned out to have a plastic body conforming to pretty much four standard styles - the squarish-cuboid, elongated-cuboid, flat-bar and cylindrical. The body being made primarily of plastic raises some concern as these are pretty heavy objects. These days even the ultra light phones don't always survive a standard drop test so heavier objects are at an obvious disadvantage. What would have been better is if they had some sort of rubber padding to absorb part of the impact shock. The Cooler Master Power Fort series and the AviiQ AV-PBPI-46RD came through on this point

READY RECKONER

There are times when you simply couldn't care for all the numbers and graphs. All you want is 3-4 options to choose from, which is why we've put together a ready reckoner featuring products that adhere to one or two factors, read on to figure out which one fits your need.

Highest capacity

All of these units should be able to provide 2-3 charge cycles for your device.

1. CyberPower CPBC10400
2. PNY Power-80A
3. Zebronics ZEB-PG7200
4. Cooler Master PowerFort 6600
5. Cooler Master PowerFort 5600 / Artis PB 5600
6. Eveready UM52

Most capacity per rupee

If squeezing out the most capacity per buck is what you are after then the following products should do the trick.

1. Zebronics PG7200
2. Portronics PICO II 2200
3. Zebronics ZEB-PG4800

Most capacity per unit volume

If storage space is a concern then the first three products offer the most capacity per unit volume, however, they are quite bulky so the next three are flat ones that easily fit in your pockets.

1. Portronics Charge X
2. Artis PB 5600
3. PNY Power 80-A

4. AsiaPower AP-4000A
5. Luxa2 P3-2500
6. AviiQ AV-PBPI-46RD

Great build quality

These models feature a more rugged and composite build.

1. Cooler Master PowerFort 6600 & 5600
2. AviiQ AV-PBPI-46RD
3. Asia Power AP2200B

Choice of colour

When you simply don't care whether it is heavy or expensive and the only thing that matters is that you want in red, yellow, green or pink then you can check out the power banks from Artis and Cooler Master.